

Goldman Sachs ELS Technical Challenge: Mutual Funds Calculator

Group 18

Table of contents

01

Overview

02

Demo

03

Backend

04

Frontend

05

Bonus features

06

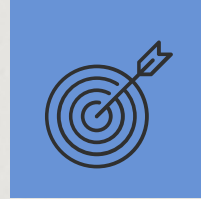
Conclusion



01

Overview

Objectives



Our aim

Create a website where you can predict the future value of an investment

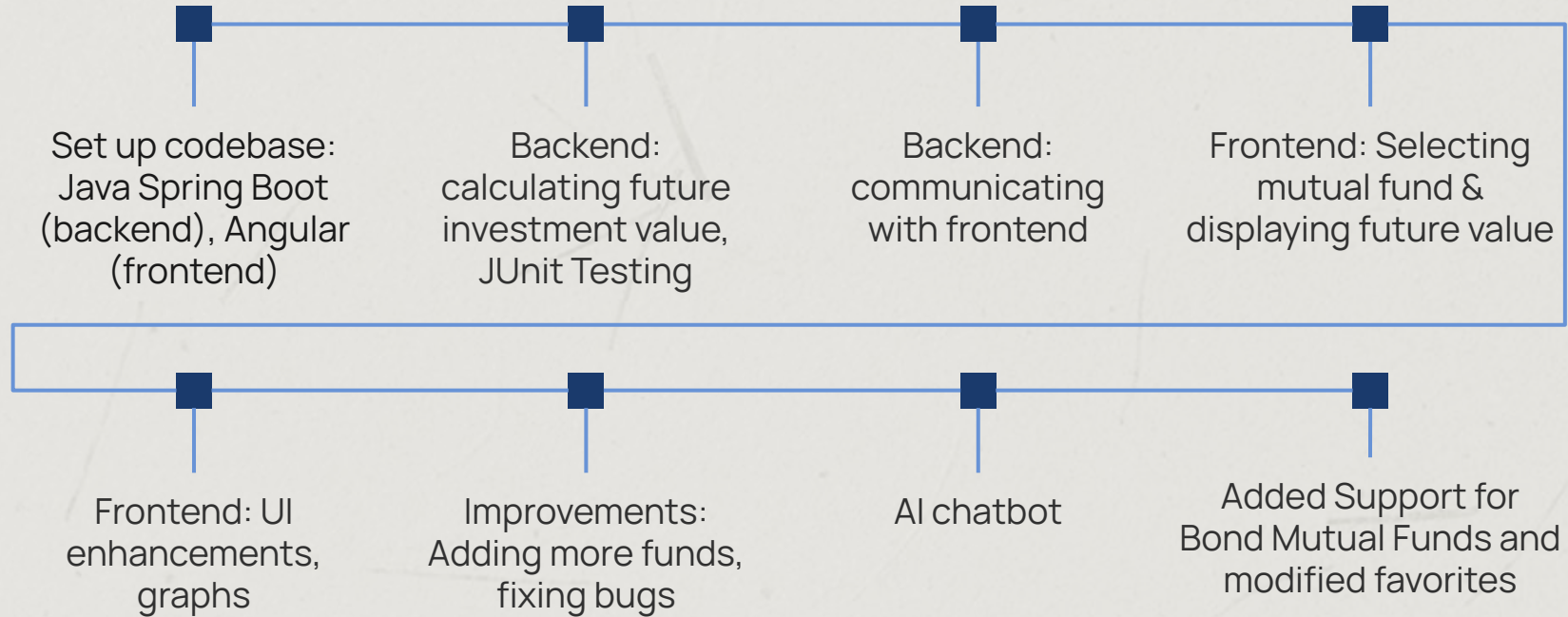
$$FV = Pe^{rt} \quad R_i = \alpha + \beta_1(\text{Term Factor}) + \beta_2(\text{Credit Factor}) + \epsilon$$

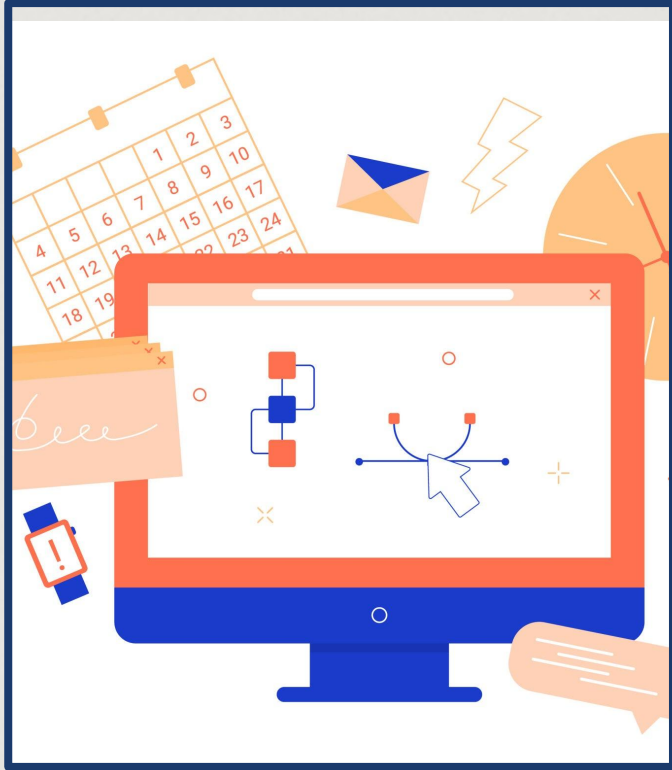
P: principal (initial investment)

t: time

r = rate = risk free rate + β * (expected return rate - risk free rate)

Project timeline





02

Demo

Demo

0	\$0.000.00	\$0.000.00	-\$0.000.00	-0.00%
1	\$0.000.00	\$0.000.00	-\$0.000.00	-0.00%
2	\$0.000.00	\$0.000.00	-\$0.000.00	-0.00%
3	\$0.000.00	\$0.042.27	-\$0.042.27	-0.04%
4	\$0.000.00	\$0.087.04	-\$0.087.04	-0.09%
5	\$0.000.00	\$0.139.30	-\$0.139.30	-0.14%
6	\$0.000.00	\$0.199.79	-\$0.199.79	-0.20%
7	\$0.000.00	\$0.269.44	-\$0.269.44	-0.27%
8	\$0.000.00	\$0.349.81	-\$0.349.81	-0.35%
9	\$0.000.00	\$0.439.88	-\$0.439.88	-0.44%
10	\$0.000.00	\$0.539.30	-\$0.539.30	-0.54%
11	\$0.000.00	\$0.649.04	-\$0.649.04	-0.65%
12	\$0.000.00	\$0.769.43	-\$0.769.43	-0.77%
13	\$0.000.00	\$0.899.48	-\$0.899.48	-0.90%
14	\$0.000.00	\$1.039.29	-\$1.039.29	-1.04%
15	\$0.000.00	\$1.189.00	-\$1.189.00	-1.19%
16	\$0.000.00	\$1.349.00	-\$1.349.00	-1.35%
17	\$0.000.00	\$1.519.00	-\$1.519.00	-1.52%
18	\$0.000.00	\$1.699.00	-\$1.699.00	-1.70%
19	\$0.000.00	\$1.889.00	-\$1.889.00	-1.89%
20	\$0.000.00	\$2.089.00	-\$2.089.00	-2.09%





03

Backend

Overview

Framework: Java Spring Boot, REST over HTTP

Database: Hardcoded mutual funds, Postgres for bond funds

MutualFund Class: ticker, name, risk, alpha, beta1, beta2

1. Initialization Layer

- Populate Postgres database with mutual funds and their alpha, beta1, beta2 values

2. Controller Layer

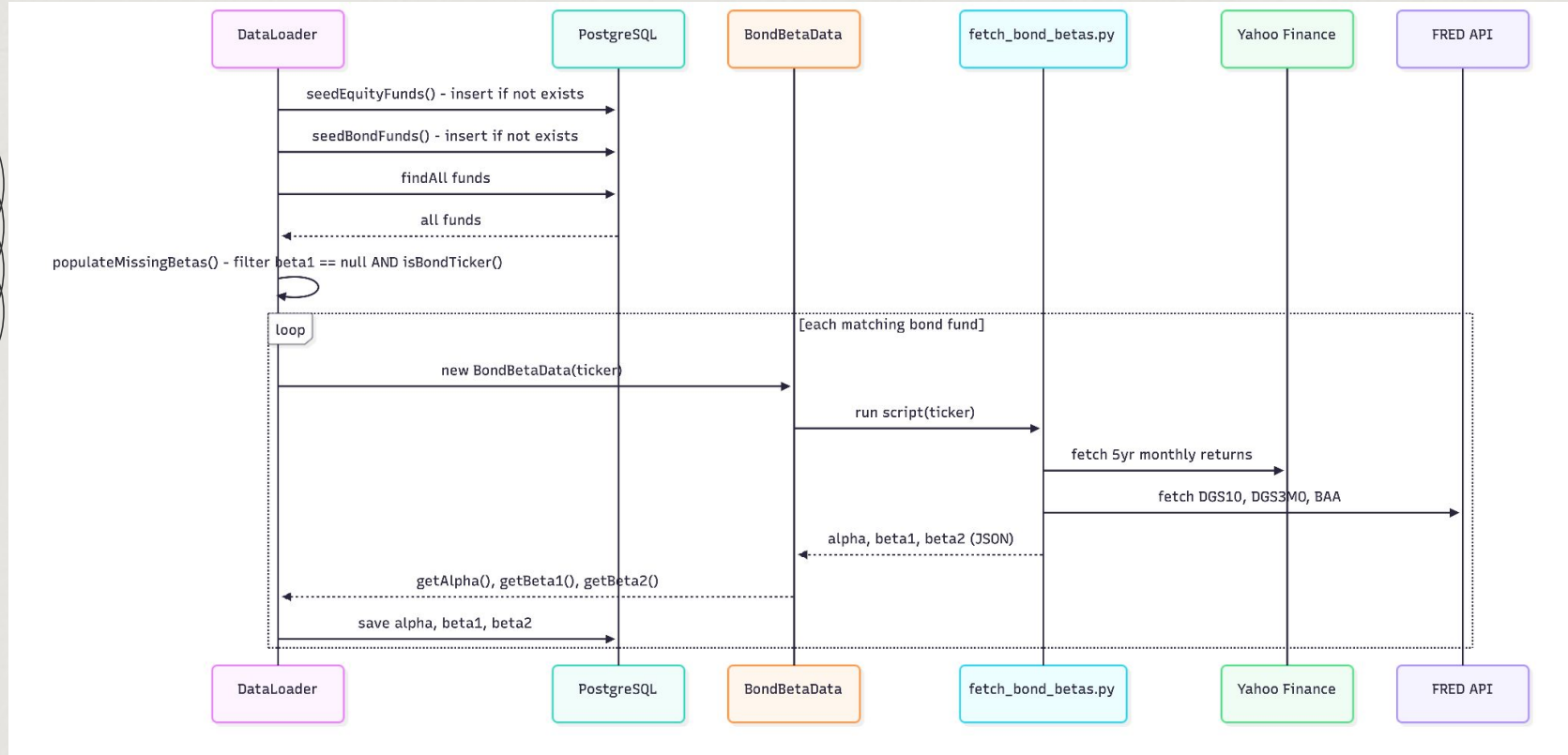
- communicate with the frontend
 - i. GET /api/funds → returns list of mutual funds and bond funds
 - ii. GET /futureValAll → calls Services to calculate projected values
 - iii. POST /api/chat → calls Services to get Gemini response

Overview

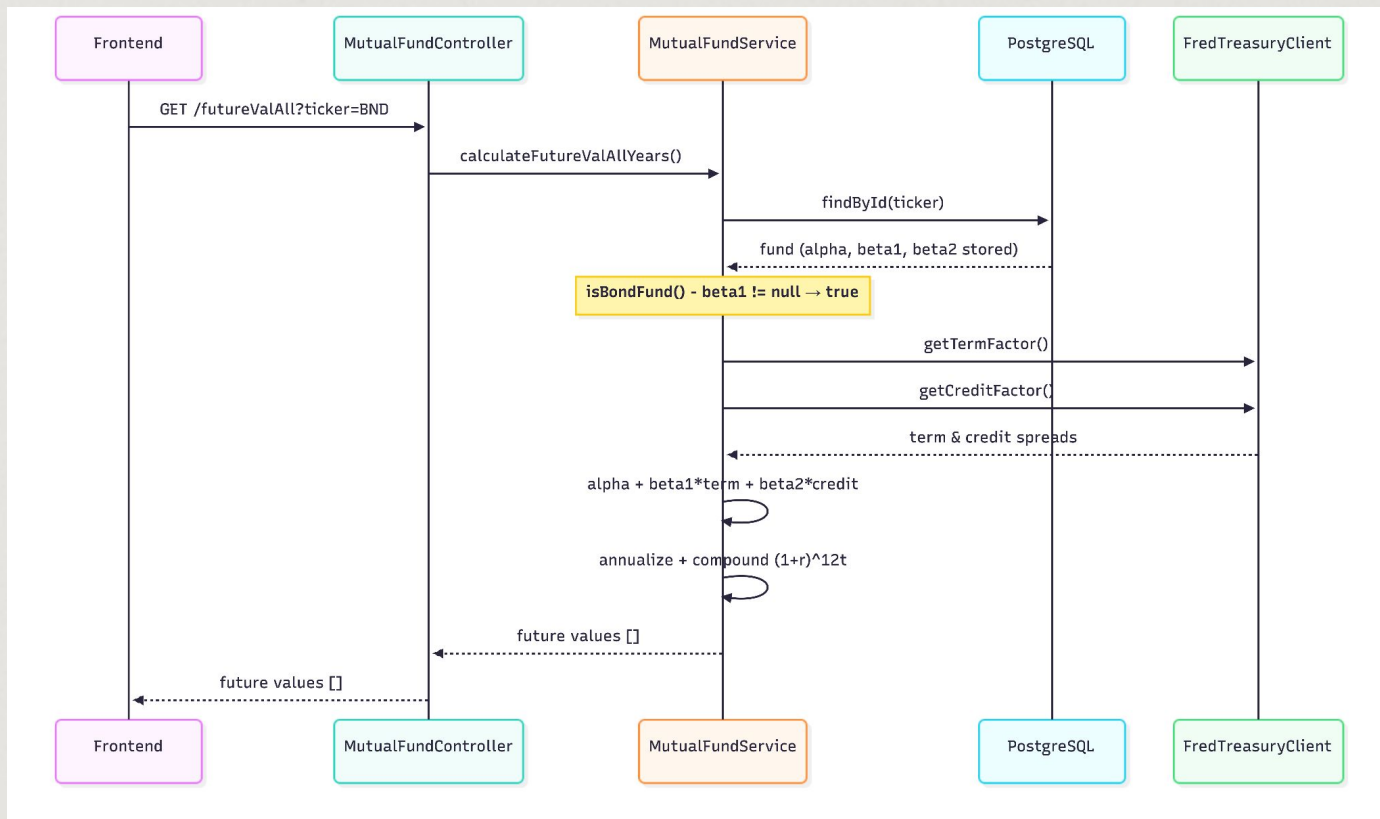
2. Services Layer

- a. Get Newton Beta value from Newton Analytics API and risk-free rate from FRED Treasury Yield
- b. Calculate expected return rate
 - i. Mutual funds: expected annual return from past five years from Yahoo Finance
 - ii. Bond funds: annual return with Term and Credit factors from FRED, alpha/beta1/beta2 from Postgres
- c. Calculate future investment value

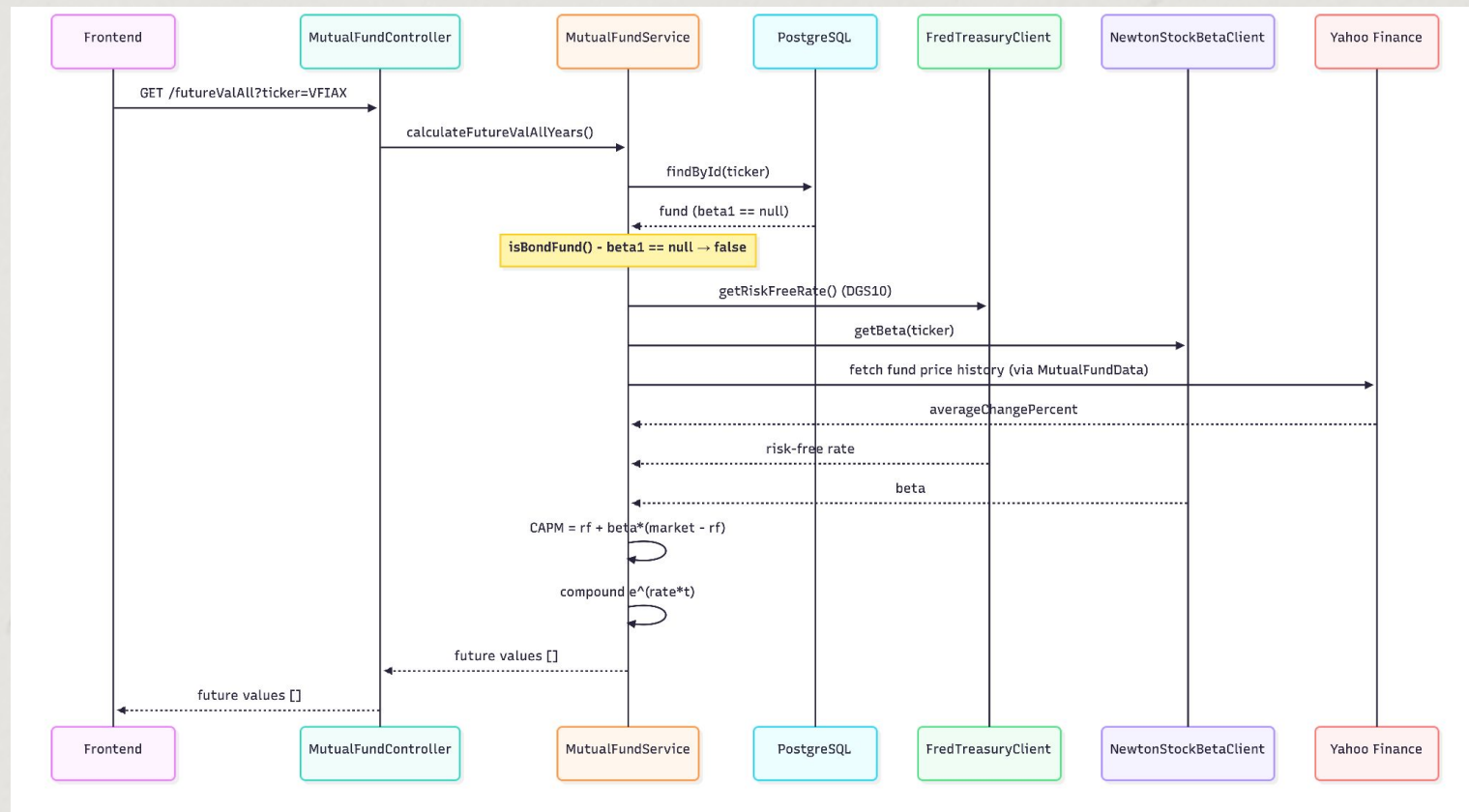
Startup Sequence



Bond Request Flow



Equities Request Flow





04

Frontend

Initial Glance

Mutual Fund Calculator

Estimate the future value of your investment

Enter Your Investment Details

SELECT A FUND

Choose a fund

HOW MUCH ARE YOU INVESTING TODAY? (\$)

10000

HOW MANY YEARS WILL YOU STAY INVESTED?

5 years

Calculate Future Value

Fund Selection

SELECT A FUND

Choose a fund

Search by name or ticker...

FAVORITES

Low Risk (Bond Funds)

FZILX Fidelity ZERO International Index Fund

PRDGX T. Rowe Price Dividend Growth Fund

Medium Risk (S&P 500)

VFIAX Vanguard 500 Index Fund Admiral

FXAIX Fidelity 500 Index Fund

High Risk (Growth / Small Cap)

VSMAX Vanguard Small-Cap Index Admiral

Dropdown, sorted by
categories

Fund Selection

SELECT A FUND

Choose a fund

Fidelity

SEARCH RESULTS

FZILX Fidelity ZERO International Index Fund

FXAIX Fidelity 500 Index Fund

FCNTX Fidelity Contrafund

FDGRX Fidelity Growth Company Fund

FSKAX Fidelity Total Market Index Fund

FMCSX Fidelity Mid-Cap Stock Fund

FICEV Fidelity International Growth Fund

Search filter for funds

Fund Selection

SELECT A FUND

PRDGX – T. Rowe Price Dividend Growth Fund



Low Risk · Steady, modest returns. Lower chance of loss.

FXAIX – Fidelity 500 Index Fund

Medium Risk · Historically strong long-term returns.

SWLGX – Schwab Large Cap Growth Fund

High Risk · Higher potential returns, more volatile.

Details about selected
fund and risk level

Principal Amount

HOW MUCH ARE YOU INVESTING TODAY? (\$)

9000

Text box for initial investment amount

- Edge case handling:
 - can't input letters
 - negative numbers not accepted

Please enter a valid investment amount.

Year Selection

HOW MANY YEARS WILL YOU STAY INVESTED?

5 years

1 year

2 years

3 years

5 years

7 years

10 years

15 years

20 years

Dropdown selection of
years

Results

Results for FZILX

YOU INVEST \$10,000.00	ESTIMATED PROFIT AFTER 5 YEARS +\$1,776.72	ESTIMATED FUTURE VALUE \$11,776.72
---	---	---

Based on historical 5-year average annual return (trading days only, holidays excluded). Total growth: +17.8%

Outputs calculated estimated future value and profit after selected number of years

Error handling for unforeseen errors (e.g., user does not have yfinance installed)

Unable to calculate future value.

Results

Growth Year by Year



Year-by-year growth
and breakdown

Year-by-Year Breakdown

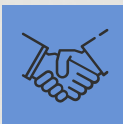
YEAR	AMOUNT INVESTED	ESTIMATED VALUE	ESTIMATED PROFIT	TOTAL GROWTH
1	\$10,000.00	\$10,332.49	+\$332.49	+3.3%
2	\$10,000.00	\$10,676.03	+\$676.03	+6.8%
3	\$10,000.00	\$11,030.99	+\$1,030.99	+10.3%
4	\$10,000.00	\$11,397.76	+\$1,397.76	+14.0%
5	\$10,000.00	\$11,776.72	+\$1,776.72	+17.8%



05

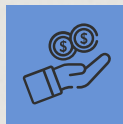
Bonus Features

Bonus Features



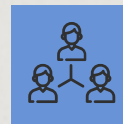
JUnit Testing

Testing the backend



UI Enhancements

Graph visualization of
investments



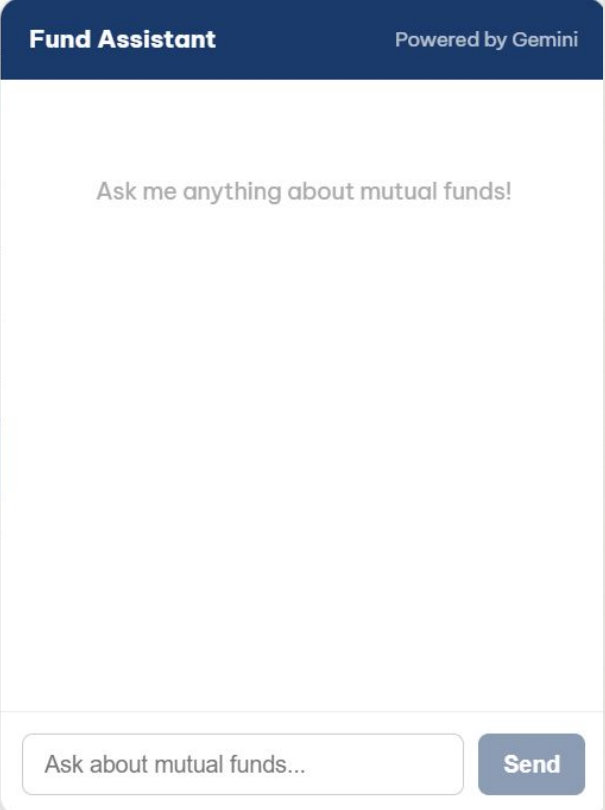
AI Challenge

Chatbot using Gemini



AI Chatbot

- Only recommends funds in our list
- Fetches live beta for funds mentioned by name by user
- Uses tool calling to fetch live expected return and future value projections
- Defined rules for recommendations, risk explanations, and budget constraints



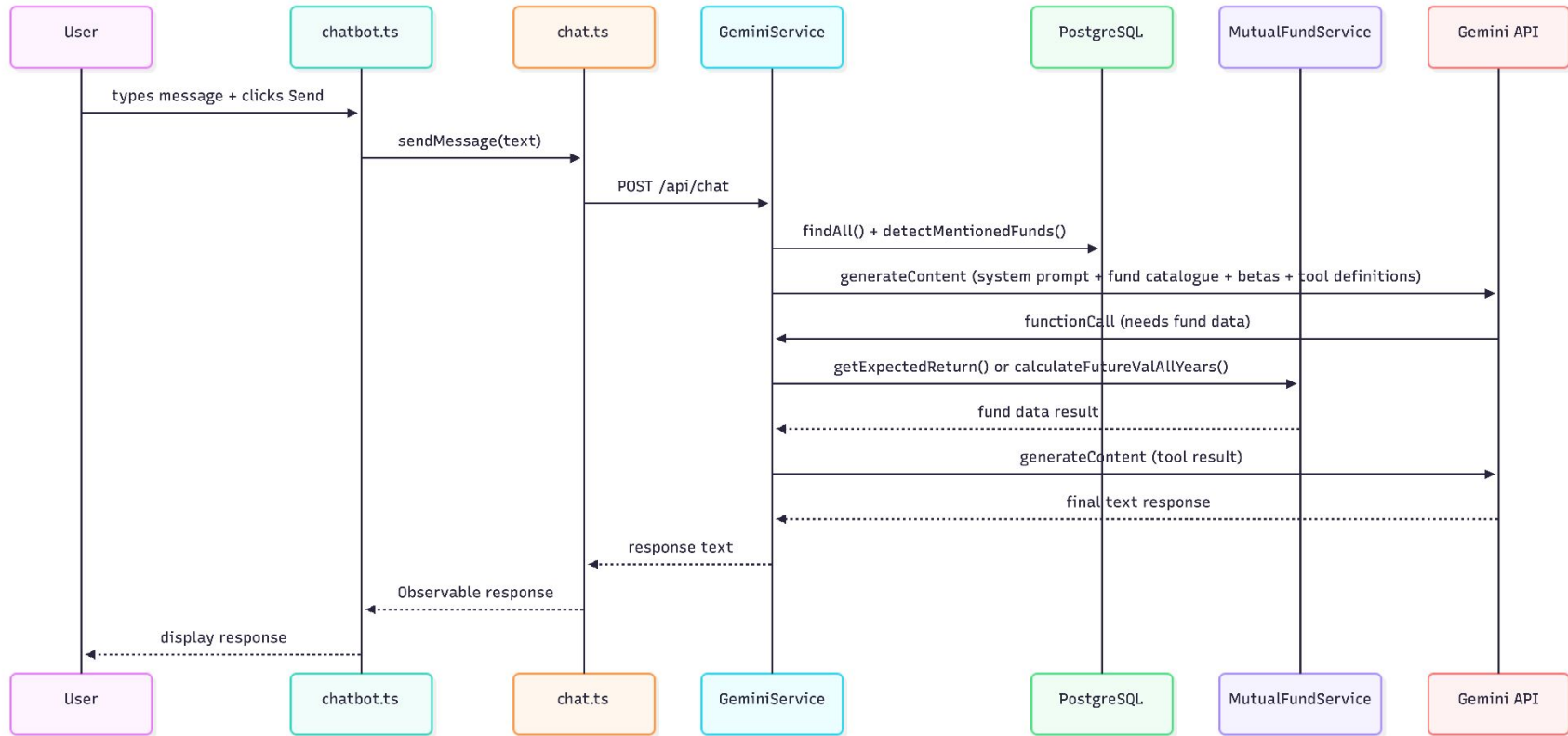
Fund Assistant Powered by Gemini

Ask me anything about mutual funds!

Ask about mutual funds... Send



Gemini API Sequence



Challenges

1

Adopting
TypeScript/Angular UI
Framework

2

Calculating for bond
mutual funds using Fama
French Model

3

Coordinating with others
with different conflicting
schedules

4

Time limitations

5

Long distance
collaboration



06

Conclusion



Conclusion

Collaboration

Division of tasks and meeting outside of scheduled ELS sessions

Backend

Spring Boot, RESTful APIs

Frontend

Angular, TypeScript

Project Management

Setting a timeline, coming up with additional features

Guest Lectures

Insights from industry professionals that shaped our approach to the project and the work ethic we brought to it

Future Direction

- Allowing users to compare multiple funds side by side
- Adding portfolio allocation to let users split an investment across several funds





Thanks!

Special thanks to Faith and Zane for helping us and giving us advice :)

Questions?

CREDITS: This presentation template was created by [Slidesgo](#), and includes icons by [Flaticon](#), and infographics & images by [Freepik](#)





07

Appendix

MutualFund Class

Java Class with three private variables

1. Ticker
 - a. Identifies the fund when calculating the future investment value
2. Name
 - a. Actual full name of the fund
3. Risk
 - a. Give more information about the fund if the user wants to select a fund with a specific risk
4. alpha, beta1, beta2
 - a. for bond funds

Initialization Layer

1. DataLoader.java
 - a. fetch_bond_betas.py
 - i. gets monthly returns from Yahoo Finance and calculates bond fund regression coefficients (alpha, beta1, beta2)
 - b. BondBetaData
 - i. gets alpha, beta1, beta2 from the JSON
 - c. seeds Postgres database with bonds and the alpha/beta1/beta2 values for each

Controller Layer

MutualFundController: Communicates with the frontend

GET /api/funds

- Returns a hard-coded list of 48 mutual funds and 20 bond funds from the postgres database
 - low, medium, high risk
 - Index funds, Small/Mid Cap funds, International funds, Sector funds

GET /api/futureValAll

- calls MutualFundService to calculate the projected values of every year until the given year

Service Layer

GET futureValAll

1. calculate expected return rates for mutual funds
 - a. `fetch_fund_history.py`
 - i. loads past data from Yahoo Finance into a JSON object
 - ii. calculates average change of the past five one-year windows and averages those percentages
 - b. `MutualFundData.java`
 - i. gets the annual average price change from the JSON output

Service Layer

MutualFundService

- a. Get rate of ticker
 - i. Cache the rate so it's not recalculated each time
- b. Get mutual fund expected return rate from MutualFundData
- c. Calculate bonds expected rate with term and credit factors, alpha, beta1, beta2
 - i. Get term and credit factors from FredTreasuryClient.java
- d. Calculate future value
 - i. Fetch beta from NewtonStockBetaClient.java (Newton Analytics API), risk free rate from FRED



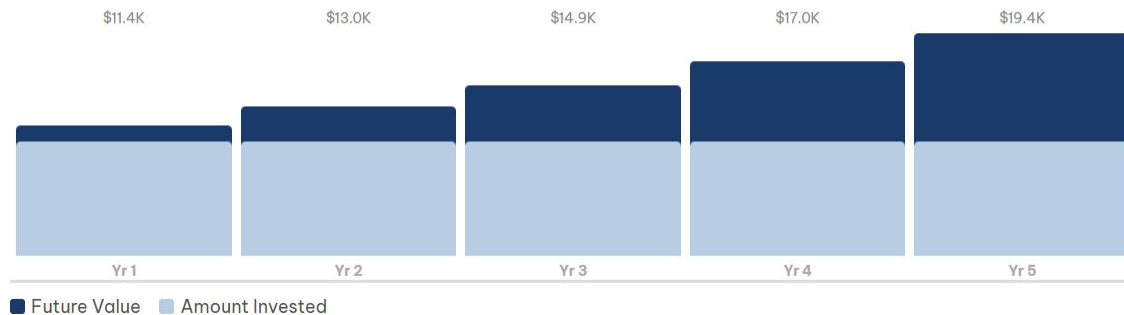
JUnit Testing

1. FutureValTest
 - a. Calls `MutualFundService.calculateFutureValAllYears()`
 - b. Asserts that the returned list has exactly 'years' entries
 - c. Asserts each future value is >0 and not NaN
2. MarketReturnTest
 - a. Constructs a new `MutualFundData` class
 - b. Calls `MutualFundData.getAverageChange()`
 - c. Asserts that the result is not infinite and not NaN



UI Enhancements

Growth Year by Year



1. Bar Graph
Visualization of
year by year
growth

2. Table

Shows estimated
value and profit of
each year as well
as total growth

Year-by-Year Breakdown

YEAR	AMOUNT INVESTED	ESTIMATED VALUE	ESTIMATED PROFIT	TOTAL GROWTH
1	\$10,000.00	\$11,422.96	+\$1,422.96	+14.2%
2	\$10,000.00	\$13,048.39	+\$3,048.39	+30.5%
3	\$10,000.00	\$14,905.12	+\$4,905.12	+49.1%
4	\$10,000.00	\$17,026.06	+\$7,026.06	+70.3%
5	\$10,000.00	\$19,448.79	+\$9,448.79	+94.5%